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Automatic Detection of Fake News Using Gated Recurrent Unit Deep Model

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Abstract

Nowadays, the utilization of social media platforms continues to rise, as is the dissemination of misleading data. This fake information is damaging to both society and individuals. Hence, it is essential to prevent and detect the dissemination of fake news. Furthermore, the main challenge is the lack of an effective technique for distinguishing between truthful and false reviews; even individuals are usually unable to recognize the difference. In contract to numerous machine learning-based methods, the deep learning-based methods exhibit a superior capacity for accurately detecting fake news. Earlier studies focused on machine learning based and data mining techniques, with limited exploration of deep learning-based methods for automated detecting fake news task. This study attempts to introduce a method based on the GRU deep model in order to detect automatically fake news. Experimental evaluation using a benchmark fake news dataset indicates extremely encouraging results and enhanced performances.

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Keywords: Fake News Detection, Text Mining, Text Classification, Deep Learning, Gated Recurrent Unit.

1. Introduction

Today, the rapid growth of social media platforms is accompanied by a surge in rumours and fake news, which can have damaging effects on both society and individuals. Therefore, it is necessary to prevent and detect the dissemination of fake information.

The topic of disinformation gained significant attention in 2016, particularly in the wake of the United States presidential elections.

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According to [1], ”rumour” and ”fake news” are two intimately interrelated terms. The term ”Fake news” or ”disinformation” has been intentionally created, whereas ”rumours” are unverified and questionable data that is circulated with no intention of misleading.

Determining the motives of those who spread information on social media can often be a challenging task. Hence, on the Internet, any incorrect or wrong information is often labelled as disinformation. Distinguishing between truthful and misleading information is considered as a challenging task. Many attempts, however, have been explored to tackle this issue. To detect misleading information transmitted online, numerous algorithms based on machine learning have been applied [2, 3, 4, 5]. Recently, deep learning is an emerging technique in the research community, demonstrating greater efficiency in detecting false information compared to traditional machine learning methods. This superiority arises from its ability to extract high-dimensional features and its reduced dependency on data pre-processing [6].

The main goal of this current study is to enhance the classification accuracy of the Fake News Detection (FND) task by means of employing various deep learning-based techniques, specifically Long Short-Term Memory (LSTM), Recurrent Neural Networks (RNN), and Gated Recurrent Unit (GRU). The research contributions of the present study can be outlined as follows:

- Proposing an efficient method for automatically distinguishing between a truthful and fake information, a task that often challenges human judgement.
- Analyzing the significance of employing a GRU deep based model in comparison to other deep learning-based methods for FND task.
- Attaining a remarkable classification accuracy of 99.83%, making a groundbreaking achievement in the arena of automated detecting fake information.

The rest of this paper is organized as follows. In section 2, we offer an overview of recent research related to methods for detecting fake news. Section 3 outlines our proposal. In section 4, we provide our experiments, including the gained results, followed by a discussion of the predictive outcomes. Lastly, in section 5, we summarize the paper and suggest potential directions for future research.

2. Related Work

The present section offers an overview of the recent research related to FND-based methods by means of employing various deep learning techniques.

In the work of [7], the authors proposed a deep CNN based model. It employs several hidden layers in order to acquire discriminatory features towards the FND task. The experimentation achieved a test accuracy of 98.36%. Although their proposed model can be less prone to overfitting issue, it requires longer to be trained.

A based Bidirectional Graph Convolutional (Bi-GCN) based model is proposed in [8]. Their proposed was for the purpose of studying characteristics through focusing on both top-down GCN and bottom-up GCN rumor propagation.

A multi-task learning model designed for FND task is presented in the work of [9], in which they examine the effect of topic categories on fake news task and leverages contextual news information in order to enhance the performance when dealing with short fake news. Their model comprises two components, including representation learning as long as multi-task learning. These components are simultaneously train two subtasks: news topic classification and FND. Nevertheless, without the author’s information, the model’s performance decreases.

In the study of [10], they proposed a hybrid based architecture, that leverages the strengths of both LSTM and CNN. Their proposed architecture was employed with two distinct dimensionality reduction techniques, namely Chi-Square and Principal Component Analysis (PCA). Their work involved the acquisition of a dataset gathered from ”Fake News Challenges (FNC)” website, comprising four kinds of stances: ”agree”, ”disagree”, ”discuss”, and ”unrelated”. Their model achieved a remarkable classification accuracy of 97.8%. Furthermore, their results indicated that PCA outperformed Chi-square as a dimensionality reduction technique.

The employment of machine learning based methods for the task of FND, specifically supervised learning, was proposed by [11]. They employed a dataset containing both false and truthful news articles for training a machine learning-based model, implementing by means of Scikit-learn library for this purpose. Feature extraction from the used dataset was achieved through text representation-based techniques, including Bag-of-Words (BoW), Bi-gram

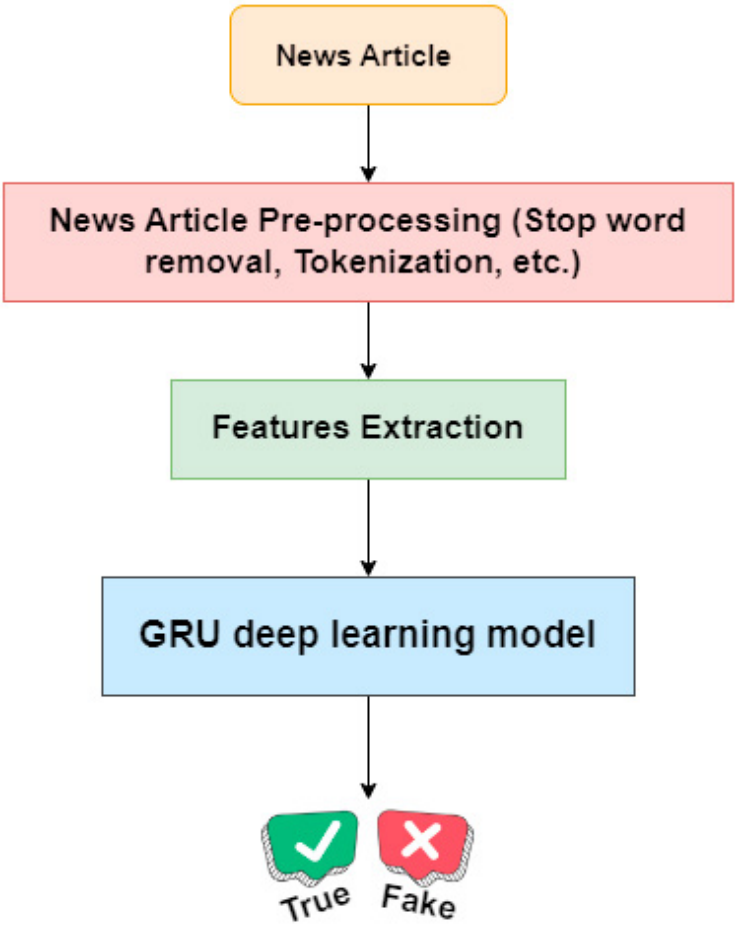


Fig. 1. An overview of the main structure of our proposal for FND task.

frequency, and Term Frequency-Inverse Document Frequency (TF-IDF) representations. It is noteworthy that the size of the feature vectors employed had a significant impact on the classification performance of their proposed model.

In a study conducted by [12], they introduced an end-to-end system named the ”event adversarial neural network”. This system is designed in order to extract and identify event-invariant characteristics, thereby facilitating the identification of fake information, especially in the context of incoming events. The dataset utilized for their research was collected from both Weibo and Twitter. Furthermore, another study, conducted by [13], also focused on the aforementioned sources of information.

3. Proposed Architecture

The goal of this section is to present our FND architecture for English articles, consisting of two main modules: a News article pre-processing component and a fake news classification component based on a GRU deep model. Fig. 1 provides an overview of the main structure of our proposal for the FND task.

3.1. News article pre-processing

Before classifying articles into their respective labels, it is necessary to subject the data to specific refinements, including stop word and punctuation removal, tokenization, sentence segmentation, lowercasing. These preprocessing

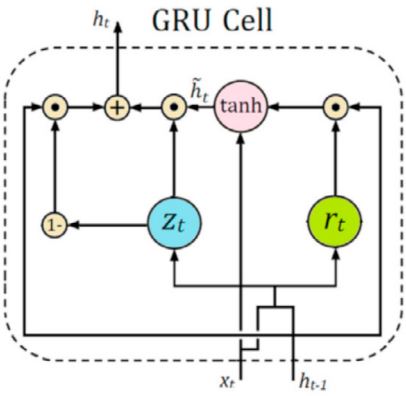


Fig. 2. The GRU cell [15].

steps are crucial for reducing the overall size of the dataset by eliminating irrelevant information. To achieve this, we developed a generic preprocessing function. This function first removes punctuation and non-letter characters from each article. Following this, it converts all letters to lowercase. Additionally, we employed the tokenizer[1] from the popular Keras library to transform the text in input articles into sequences of integers.

3.2. Fake news classification module based on a GRU neural network

In our study, we leverage the GRU deep based model as a powerful technique for fake news classification. The model's ability to capture long-term dependencies, learn informative features, and mitigate overfitting makes it well-suited for this task. GRU features gating mechanisms that enable it to regulate the flow of information through the network. These include an update gate r, which controls the retention of information from the previous time step, and a reset gate z, which determines how much new information to incorporate. These gates are instrumental in capturing and preserving relevant information while discarding noise [14] (Fig. 2). Within the GRU architecture, the transition process was executed through the utilization of the subsequent equations:

$$r_t = \sigma(W_r * x_t + U_r * h_{t-1} + b_r \tag{1}$$

$$z_t = \sigma(W_z * x_t + U_z * h_{t-1} + b_z \tag{2}$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \widetilde{h}_t \tag{3}$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \widetilde{h}_t \tag{4}$$

4. Experiments, Results, and Discussion

This section assesses the efficiency of our proposed FND-based model by employing a benchmark ISOT Fake News dataset. The first section introduces the benchmark used dataset and the second section presents and deliberate the experimental results.

4.1. ISOT English Fake News Dataset

In this study, we have used the benchmark dataset called ISOT Fake News Dataset [2, 3]. The dataset was available online[2]. The used dataset includes two distinct categories of articles: "Fake News" and "Real News". This dataset

[1] tf.keras.preprocessing.text.Tokenizer — TensorFlow v2.13.0', TensorFlow
[2] Fake News Detection Datasets — ISOT research lab

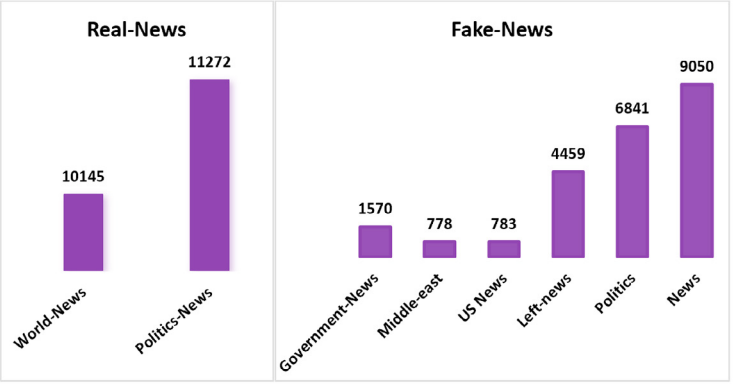


Fig. 3. The statistical information of the used ISOT Fake News dataset [2, 3].

was ultimately assembled from authentic real-world resources; where the correct articles were gathered from the website Reuters.com, while the fake articles were gathered from various kinds of sources. We notice that the "Fake News" was gathered from unreliable websites that had been reported through Wikipedia and "Politifact" (a US-based fact-checking organization). The used dataset covers several kinds of articles on various subjects, having most of the articles concentrating on political and world news. Fig. 3 displays the statistical information of the used ISOT Fake News dataset.

4.2. Experimental Results and Discussion

The aim of this section is to provide our experimental results. Firstly, we compared the performance of our proposal with two widely used deep learning techniques on the benchmark ISOT fake news dataset in term of classification accuracy and AUC [16] (Experiment 1). Secondly, we illustrate both the model accuracy and model loss on both training and testing data, aiming to demonstrate any overfitting concerns in the proposed deep-based model (Experiment 2). Finally, we showcase a confusion matrix to provide a comprehensive perspective on the classification performance of our proposal (Experiment 3).

Experiment 1: Accuracy and AUC

Within this subsection, we present an empirical analysis aimed at assessing the significance of utilizing the proposed GRU-based deep model for addressing the fake news task in comparison to other deep learning models such as LSTM and RNN. For this regard, Table 1 shows the results in terms of classification accuracy and AUC on the benchmark fake news dataset. We can observe clearly that our proposed GRU deep based model has accomplished a groundbreaking achievement in the field detecting fake news, achieving a classification accuracy of 99.83%. This can be explained through that GRU deep learning model is known to perform well in text classification tasks due to its specific architectural features that address some of the challenges inherent in processing sequential data like text. Theoretically, GRU is designed to mitigate the vanishing gradient problem by incorporating gating mechanisms that allow them to retain information over longer sequences. This ability to capture long-term dependencies is crucial in understanding the context and semantics of text data. In addition, GRU have three essential gates that can help GRU to focus on relevant information and ignore noise, aiding in better feature extraction. Furthermore, the architecture of GRU allows for efficient training compared to more complex models like LSTM.

Experiment 2: Model accuracy and Model Loss

Observing the accuracy and loss in testing/training holds the second highest importance in the process of training a classification technique. Fig. 4 and Fig. 5 provide valuable insights into the amount of overfitting observed in the novel deep-based models being proposed for fake news task. It is evident that our GRU-based deep learning approach

Table 1. The results of our proposed deep models for FND task in terms of classification accuracy and AUC.

	Accuracy	**AUC**
LSTM	0.9969	1.00
RNN	0.7448	0.8140
Our proposal (GRU deep based model)	**0.9983**	0.9998

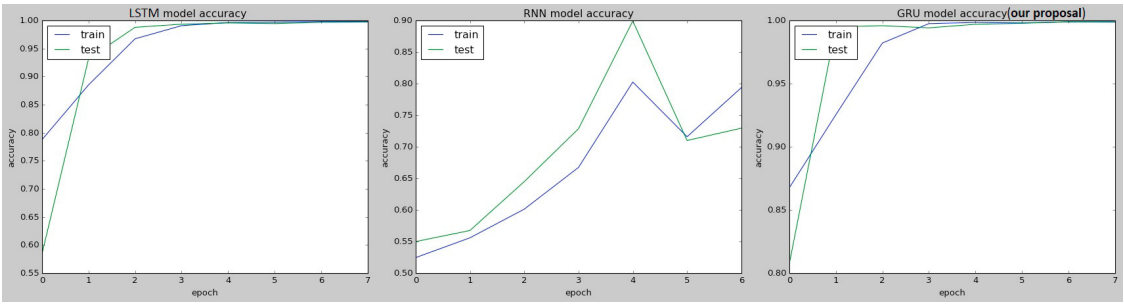


Fig. 4. Model accuracy plots of our deep learning-based models on Training and Testing data.

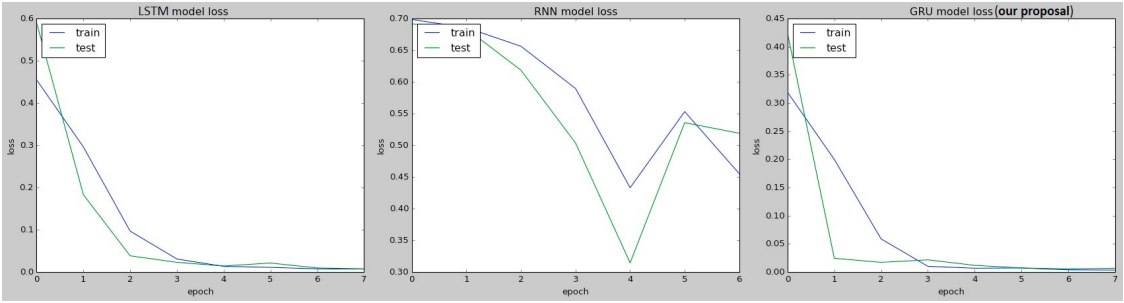


Fig. 5. Model loss plots of our deep learning-based models on training and testing data.

underwent training without experiencing any issues related to overfitting.

Experiment 3: Confusion matrix

In order to assess the quality of our model's predictions and understand how well it is performing across various articles label. Fig. 6 show the confusion matrix of the proposed deep learning-based models as fake news classification technique on the benchmark ISOT Fake News dataset. We notice that the classifier based on GRU deep learning algorithm can distinguish amongst articles label in most cases, since 0.53 real articles correctly predicted as real ones by the proposed GRU based model, 0.47 fake articles correctly predicted as fake ones, 0.00045 fake articles predicted as belonging to real articles, and 0.0012 real articles predicted as belonging to fake ones. Thus, given the relatively higher number of "true positive" and "true negative" values in the used benchmark dataset, this turned out to be an extremely efficient classification technique for fake news task.

5. Conclusion and Future Work

The GRU deep model represents a crucial advancement in the field of fake news classification. Its inherent strengths in sequential data analysis, feature learning, and overfitting mitigation contribute to enhanced classification accuracy.

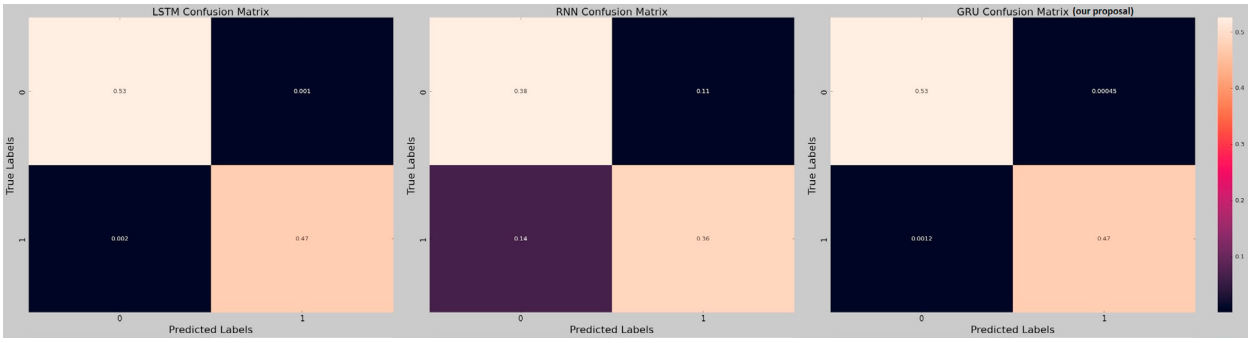


Fig. 6. Confusion matrix of the proposed deep based models.

As demonstrated in our experiments, the GRU deep model holds great promise in addressing the challenges posed by FND in today's information landscape.

In future work, we plan to propose an automated system for detecting fake information in Arabic language. We will also analyze the impact of our proposal on the tailored language. Arabic language presents several challenges, including ambiguity, agglutination, dialectal variation, and morphological richness, which necessitate a specialized investigation of our approach.

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